

Accelerometer-based HUD Input for Car Navigation

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Abstract— The current technological trend indicates a convergence of augmented reality (AR) with heads-up display (HUD) in car navigation UI's. In support of this trend, we propose use of small devices which have accelerometers or accelerometer-equipped smartphone as a basis of HUD interaction. In this paper, we defined the limited operation, experimented the basic operation and evaluated them. Then we obtained the good results.

I. INTRODUCTION

Modern car navigation systems help to convey a vehicle's current location on the map, route guidance, points of interest (POI), etc. These systems can generally be divided into three types (See Figure 1):

1. The turn-by-turn type, with no display.
2. The embedded display type, installed in the dashboard or carried by the user.
3. The HUD type, which visualizes information on the vehicle's windscreen.

Although the HUD type is the most expensive, and remains somewhat limited in both input and output, it has the distinct advantage of minimizing driver distraction. To support the further development of HUD-based car navigation, we propose a new kind of HUD system that integrates a smart mobile device with an accelerometer.

II. REQUIREMENTS

Our system assumes the existence of a vehicular HUD with

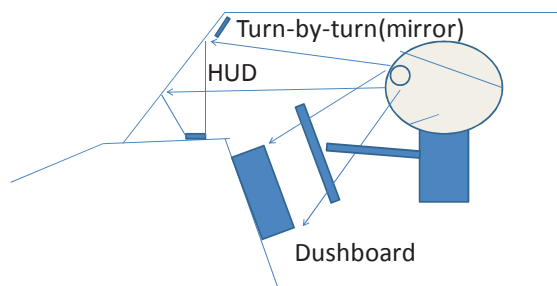


Figure 1. Three types of navigation systems

the following characteristics:

1. A simple user interface with intuitive operations
2. No-touch interaction (the user can only watch what is displayed on the windscreen)

To interact with the HUD, we proposed use of a mobile “smart” device equipped with touchscreen and accelerometer, communicating with navigation systems via Bluetooth or WiFi.

III. RELATED STUDIES

There are wireless mouse technologies. However, these interfaces are different from the smartphone, and users have to bring the mouse devices.

Interfaces for wearable computing [1] have shown promise, but are ill-suited to driving, since drivers should not have to wear anything. Image processing technology with a camera is more suitable for this purpose, but requires intensive analysis of various positions under unpredictable lighting conditions.

Accelerometer-based input systems [2] [3] are generally simple, but poorly suited to controlling the position of a pointing marker. However, since our target system involves a HUD, we can constrain the pointing range of our accelerometer input to the display area of the windscreen.

IV. PROPOSED METHOD

A. Overview

We base our system on two types of input operations:

1. Tap operations—indicating selection actions
2. Double-tap operations—indicating zooming actions

Given the limited information available on most HUDs, these two operations are sufficient for navigation interactions. To support such operations, we propose use of an accelerometer device for pointing. As with many mouse-based pointing interfaces, the color of the item beneath the pointer should be altered.

B. Structure of proposed system

Our system (See Figure 2) has three components:

1. HUD
2. Vehicle Information Device

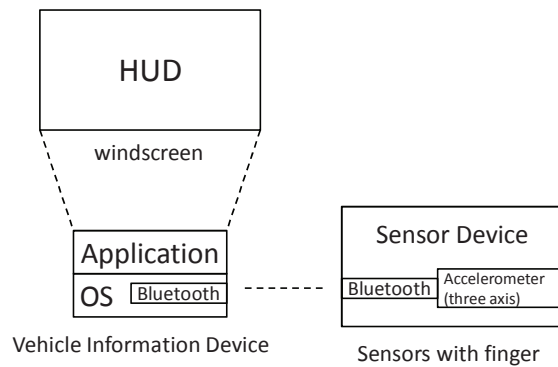


Figure 2. Structure of proposed system

3. Sensor device

The Vehicle information device is connected to the sensor device via Bluetooth. The sensor device sends raw input data from its 3-axis accelerometer. The vehicle information device processes this input and performs the appropriate operations.

V. BASIC EXPERIMENTATION AND EVALUATION

We performed an experiment using a sensor device mounted on the back of a glove (see Figure 3). Figures 4 and 5 show the accelerometer's vector input for tap and double-tap operations, respectively. Note that the double-tap operations can be clearly distinguished from tap operations by counting the number of peaks (i.e. values above a given threshold) within fixed period. Figure 6 shows the input for long tap operations. Note that it is comparatively difficult to distinguish from the tap operation, given its similarity to the tap operation.

VI. CONCLUSION

In this paper, we proposed a user interface for HUD-based car navigation. Experiments showed that the distinctness of accelerometer input was sufficient for the limited operations of most HUD-based systems.

In the future work, we plan to implement a fully functional car navigation system, and to test other factors including driver distraction, alternate pointing techniques, etc.

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Figure3. Basic experimentation by accelerometer on the back of a hand

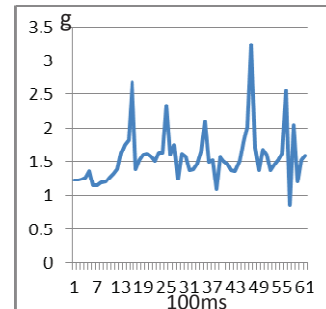


Figure 4. Value of the tap operation

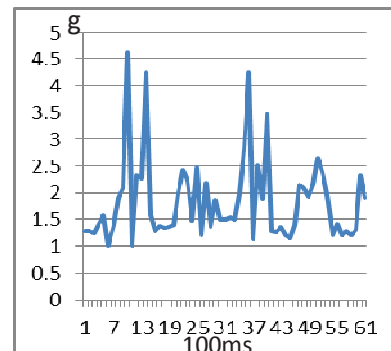


Figure 5. Value of the double tap operation

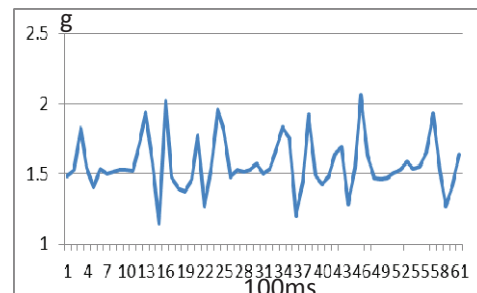


Figure 6. Value of the long tap operation